A narrative introduction to probability and Bayesian theory
BMI/COMPSCI/PSYCH 841: Computational Cognitive Science
Prof. Austerweil[1]
Jan 21 2025 Edition

1. **Basic probability theory**. A *set* is a collection of elements or members. Sets are defined by the elements they do or do not contain. In the beginning, there is a set or space of possible outcomes (often denoted as Ω). At least one outcome must be possible, so the *set size* of an outcome set Ω is greater than or equal to 1. To be fancy, people often write this as $|\Omega| \geq 1$ (instead of size, you might hear it referred to as *cardinality* for extra fanciness). For example, Ω could be the result of a coin flip ($|\Omega| = |\{H, T\}| = 2$), the result of 2 coin flips ($|\Omega| = |\{HH, HT, TH, TT\}| = 4$), the result of a coin flip and what Joe decided to have for dinner ($|\Omega| = |\{\{H, \text{Bacon}\}, \{T, \text{Bacon}\}, \{H, \text{Ice Cream}\}, \{T, \text{Ice Cream}\}\}| = 4$, etc.[2]

2. We still don't have probabilities yet. To get a probability, you must define events. An event is a set that contains outcomes (it can be none, some, or all of the outcomes). If each outcome is equally likely, there is a simple rule for calculating the probability of an event: the probability of an event is the number of outcomes in the event divided by the number of possible outcomes. In other words, the probability of event A is $P(A) = |A|/|\Omega|$. Imagine the set of possible outcomes is the outcome of flipping two coins. Figure 1a displays the set of possible outcomes in this example.

3. We can define random variables[3] corresponding to the outcome of each coin flip, X_1 and X_2. This means that the variable X_1 denotes the value of the first coin (a function of the outcome). It can be either heads (H) or tails (T). You hear a rumbling from above, "What is the probability that X_1 is H?" Joe's voice booms "Remember item 2". You do, and draw a red box, which is shown in Figure 1b. There are two outcomes in which X_1 is heads and four possible outcomes. Thus, $P(X_1 = H) = 2/4 = 1/2$ (or the probability that the first coin is heads is 1/2).

(a) Set of possible outcomes for the result of two coin flips.

(b) Probability of an event is the ratio of the number of outcomes that have the event to the total number of possible outcomes. For example, the red box corresponds to the event "the first coin flip is H" and so, its probability is 1/2.

Figure 1: Possible outcomes and probability as the ratio of possible outcomes where an event occurs to all possible outcomes.

4. Joe gets upset with Coin 2, he says "Coin 2 has committed blasphemy and must go. Create a set of possibilities without Coin 2 and tell me the probability that Coin 1 lands heads." You comply by drawing $\{H, T\}$. You circle H and so that is one out of the two possibilities. Thus, $P(X_1 = H) = 1/2$.

---

[1] Special thanks to Kyana Burhite who provided invaluable comments and edits to improve the manuscript.

[2] Yes I only eat bacon or ice cream for dinner. I'm an adult and you can do that when you're an adult. Cf my gallbladder's eulogy.

[3] More generally, a random variable is a function that maps the set of possible outcomes (e.g., Ω) to some measurable set or space. The output or range of the function could be the set of outcomes Ω, a simple calculation (e.g., the number of heads in a space where multiple coins are flipped), or something more complex (e.g., the world's friendship matrix, an 8 billion by 8 billion, binary matrix N where $N_{1,100} = 1$ if person 1 is friends with person 100, and $N_{1,100} = 0$ otherwise). We'll move from simpler to more complex cases as we progress.

When you sum over the possible values of a variable like you did here, it is called *marginalization*. You note that the probability that Coin 1 is Heads when Coin 2 is not a possibility (marginalized probability) is the same as the probability using a random variable for Coin 1's value when Coin 2 is also a possibility.

5. Joe realizes it was his cat, Mr. Jim, who ate the last piece of bacon and not Coin 2. Thus, Joe returns Coin 2 to the set of possible outcomes. You realize that a random variable can also be an event.[4] He defines a new random variable Y, which is true when an outcome has at least one H. What is the probability of Y being true (ie., what's the probability that there is at least one heads when flipping tow coins)? In your wisdom, you note there are three outcomes with at least one heads: HT, TH, and HH. That is three of the four possible outcomes. Thus, $P(Y) = 3/4$.

6. *Conditional probability as restriction of possible outcomes.* In protest of Coin 2 being back, Coin 1 refuses to land Heads and will only land Tails. Given this knowledge, what is the probability of Coin 2 being H? This is the *conditional probability* of Coin 2 being H given Coin 1 is H or $P(X_2 = H|X_1 = T)$ (this is read as "the probability that X_2 is H given X_1 is T"). In this notation, everything to the left of the | is the event that you wish to calculate the probability of and everything to the right of the "—" is the "information being conditioned on" or the "given information". The simplest way of calculating *conditional probability* is to restrict the outcome space to those outcomes that are consistent with the given information. Thus, the new outcome space only contains those outcomes where X_1 is H ($\Omega_{X_1=H} = \{TH, TT\}$). In this new outcome space, there is one outcome consistent with the event (one where X_2 is H) of the two possible outcomes. So, the conditional probability is 1/2. Thus, conditional probability works the same as standard probability, just on a restricted subset of possible outcomes.[5] Graphically, $P(X_2 = H|X_1 = T)$ is shown in Figure 2a.

7. Coin 1 is still being stubborn. It decides always to be H now. A stranger approaches you, "If you tell me the probability that there is at least one heads, Joe will provide you with 5 points of fake extra credit." You want the fake extra credit (even if it's fake, it's still extra credit right?),[6] but how do you write the event that there is at least one H? Well, $A = \{HH, HT, TH\}$. A is now the possible outcomes that have at least one heads. With this knowledge in hand, you go back to drawing possible outcomes as illustrated in Figure 2b. The probability is 1! Joe is pleased. He says, "Fortune is with us when at least one coin lands heads." Now fortune is always with us!

8. *Dependence.* What is different between Figure 2a and 2b? In both cases we learned the outcome of the first coin flip, but in only one of them *influenced* the probability of the event (Without knowing the value of coin 1, what is the probability that there is at least one H? It is 3/4. Can you show why?). When knowing the outcome of one random variable or event influences the probability of another, those are called *dependent.* Alternatively, when conditioning on a variable does not influence the probability of another variable, these variables are *independent*. If variables X and Y are independent, it is often written as $X \perp Y$. In this case, you can drop the conditioning portion of the probability. Usually, it is good to do this when you can. Joe is lazy and likes to write less whenever possible. He recommends that you do too.

9. *Joint probability*. Coin 1 bribes Joe with \$100 and so Joe declares that Coin 1 is a good coin (it's back in the outcome space). Every value in the outcome space has equal probability (or is equiprobable).

[4] A straightforward way to do this is to define the input to be the set of possible outcomes and the output to binary depending on whether the outcome is in the event.

[5] Except it is considered undefined if you condition on an event that corresponds to no possible outcome (i.e., has zero probability). How would you calculate the conditional probability of something when you condition or take as a given an event that is impossible? To avoid strange philosophical issues here of impossible events being assumed to have happened, mathematicians declared probabilities in this strange circumstance to be undefined.

[6] No. It is not extra credit. You get no extra points for getting this correct. BUT, you might feel a bit better. Isn't that as nice as extra credit points?

(a) Conditional probability of $X_2 = H$ given $X_1 = T$. **(b)** Conditional probability of A (there being at least one H in the outcome) given $X_1 = H$.

Figure 2: Conditional probability

He asks, what is the joint probability that X_1 is H and X_2 is T. The joint probability is the number of outcomes consistent with both events divided by the total number of possible outcomes. There is 1 outcome where $X_1 = H$ and $X_2 = T$ out of 4 possible outcomes. Thus, the probability is 1/4. This is shown in Figure 3.

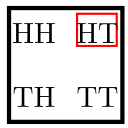

Figure 3: Joint probability of $X_1 = H$ and $X_2 = T$.

10. *More on Marginalization and Marginal Probabilities.* Intuitively, the following two ways of calculating the probability a variable takes a value should give the same answer: (1) enumerate the possible outcomes containing only that variable and count those where it has the specified value (exemplified in Figure 4a) and (2) enumerate the possible outcomes containing that variable and another variable and count all of those where the first variable has the value of interest (exemplified in Figure 4b). For example, the probability that the first coin flip is heads is 1/2 according to both Figure 4a and Figure 4b. This intuition is formalized in the *sum rule*. Consider two random variables A and B

(a) Probability that $X_1 = H$ when the set of outcomes does not include X_2 (1/2). **(b)** Probability that $X_1 = H$ when the set of outcomes includes X_2 ($2/4 = 1/2$).

Figure 4: Marginalization

that have joint probability distribution $P(A, B)$. The *sum rule* is

$$P(A) = \sum_b P(A, B = b) \tag{1}$$

11. Coin 2 is causing problems again. It decides it likes being heads more than it likes being tails. In fact, it decides it will be H 80% of the time and 20% T of the time. Although, this may feel much more difficult than what we did previously, it is really straightforward to do. Rather than each outcome having a weight of "1", the different outcomes are given different weights. We can think of what we did previously as giving each outcome equal probability, or a weight of $1/|\Omega|$. In this new world, each outcome is given the following weights: $HH = 0.8 \times 0.5 = 0.4$, $HT = \times 0.8 \times 0.5 = 0.4$, $TH = 0.2 \times 0.5 = 0.1$, and $TT = 0.2 \times 0.5 = 0.1$. Everything else works the same, but now using these weights rather than each outcome having equal weight. Joe dares you to recalculate all of the

previous calculations that we did before but with these new weights or probabilities for the outcomes. Please note whether independence or dependence relations for each circumstance still hold.

12. *Joint Probability, Conditional Probability, and Deriving Bayes' rule.* Formalizing conditional probability as conditioning to a different space of possible worlds is not the only way to define conditional probability. Another (equivalent) way is $P(A|B) = P(A,B)/P(B)$, under the assumption that $P(B) > 0$.[7] Multiplying each side of the equation by $P(B)$ yields another important connection between joint and conditional probability $P(A,B) = P(A|B)P(B)$ — the joint probability is equal to the product of first the prior probability of a variable and then conditioning on that value, the probability of the other variable. Note that by symmetry, the opposite ordering is also true: $P(A,B) = P(B|A)P(A)$. With these two results, it is straightforward to derive Bayes' rule:

$$\begin{aligned} P(A|B)P(B) &= P(A,B) = P(B|A)P(A) \\ \Rightarrow P(A|B) &= \frac{P(A,B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)} \end{aligned}$$

where we divided each side of the equation by $P(B)$. Sometimes Bayes' rule is also written in a form where we expand the possible values for B using the marginalization rule:

$$P(A=a|B) = \frac{P(B|A=a)P(A=a)}{P(B)} = \frac{P(B|A=a)P(A=a)}{\sum_{a'} P(A=a',B)} = \frac{P(B|A=a)P(A=a)}{\sum_{a'} P(B|A=a')P(A=a')} \quad (2)$$

Please note that using a' for the denominator sum is technically necessary as mathematics does not usually use a scope (or overloading) and a is already used on the left. It is often left implicit and just written as $\sum_a$ and using a rather than a' in the denominator. Also, when the author feels like it should be clear to the reader, they sometimes will drop the index from the summation notation and just write $\sum$ for the denominator. Please note that this isn't identical to when the reader feels it is clear. Curse the curse of knowledge when teaching or writing!

13. *The Taxicab Problem* (Kahneman & Tversky, 1972; Bar-Hillel, 1980). A classic problem used to evaluate human probabilistic reasoning is The Taxicab Problem:

    Two cab companies operate in a given city, the Green and the Blue (according to the color of cab they use). 85% of the cabs in the city are Green and 15% of the cabs in the city are Blue. A cab was involved in a hit-and-run at night. There was a witness. The witness accurately identifies the color of a cab at night 80% of the time. She testifies that she saw the Blue cab. What is the probability that the cab is Blue given the witness's testimony?

    Let X be the actual color of the cab involved in the hit-and-run and W be the cab color reported by the witness. The above information specifies the following quantities: $P(X=G) = 0.85, P(X=B) = 0.15, P(W=B|X=B) = 0.8, P(W=G|X=B) = 0.2, P(W=B|X=G) = 0.2$, and $P(W=G|X=G) = 0.8$ (Make sure you understand why!). We can solve for the probability that the cab is Blue given the witness's testimony, $P(X=B|W=B)$, using Bayes', marginalization and product rules.

[7] Assuming the $P(B) > 0$ is equivalent to assuming that B having at least one possible outcome in the event. If you don't understand why, please check in with a friend or Joe.

$$
\begin{aligned}
P(X=B|W=B) &= \frac{P(W=B|X=B)P(X=B)}{P(W=B)} \\
&= \frac{P(W=B|X=B)P(X=B)}{\sum_{c=B,G} P(P(X=c,W=B))} \\
&= \frac{P(W=B|X=B)P(X=B)}{\sum_{c=\{B,G\}} P(W=B|X=c)P(X=c)} \\
&= \frac{P(W=B|X=B)P(X=B)}{P(W=B|X=B)P(X=B)+P(W=B|X=G)P(X=G)} \\
&= \frac{0.8\times 0.15}{0.8\times 0.15+0.2\times 0.85} = \frac{0.12}{0.12+0.17} = \frac{0.12}{0.29} \approx 0.41
\end{aligned}
$$

Note that people typically give 0.8 as an answer (which researchers have called *Base Rate Neglect*). [8] Rather than performing the above symbol manipulation, a simpler method is to compute the product for each possibility and then normalize (this is what we ended up doing at the end!). Computing product gives us a term that is proportional to the probability. "Proportional to" is written as $\propto$. $A \propto B$ means $A = kB$ where k for some constant number.

$$
\begin{aligned}
&P(X=B|W=B) \propto P(W=B|X=B)P(X=B) = 0.8\times 0.15 = 0.12 \\
&P(X=G|W=B) \propto P(W=B|X=G)P(X=G) = 0.2\times 0.85 = 0.17 \\
&P(X=B|W=B) = 0.12/(0.12+0.17) \approx 0.41 \\
&P(X=G|W=B) = 0.17/(0.12+0.17) \approx 0.59
\end{aligned}
$$

This may be simpler for you (it is for me) and is formally equivalent. Regardless, it becomes difficult to impossible to calculate when there are a lot of different possibilities (e.g., if cabs could be any color in a continuum of colors).

14. *Continuous probabilities.* Continuous probabilities are a bit trickier. We will not define it as before, where we built a probability space on events that are subsets of outcome space. Instead, we will define probabilities over a continuous space as the area under a *probability density function*. Note that any particular x has zero probability in a continuous space. Instead, we examine the probability of a set of values under a *probability density function*. The relative values of a x_1 and x_2 according to *probability density function* $p(x)$ gives the relative likelihood of each value and $p(x)$ can be greater than 1, as long as $0 \leq \int_A p(x)dx \leq 1$ for any region A. In other words, $P(A) = \int_A p(x)dx$.

    Another tool used for manipulating continuous probabilities is the *cumulative distribution function*. The *cumulative distribution function* of a random variable X is $F_X(x) = \int_{\infty}^{x} p(t)dt$. It is useful for calculating the probability of random variable X taking a value within an interval: $P(a \leq X \leq b) = F_X(b) - F_X(a)$.

15. *Uniform.* The continuous uniform distribution places an equal probability across an interval $[a, b]$. The probability density function $p(x)$ for the continuous uniform distribution is $p(x) = \frac{1}{b-a}$. For example, if X is a continuous random variable that is uniformly distributed between 0 and 0.5 $(0 \leq X \leq 0.5)$, then $p(x) = \frac{1}{0.5-0} = 2$ (between 0 and 0.5, it is zero for other x). Note that $p(x)$ is greater than zero everywhere that it is non-zero. To signify the region that a density is non-zero,

[8] Whether the typical intuitive answer that people give to this problem is indicative of people being completely non-Bayesian when reasoning in uncertain situations or not doing it in a naive manner based solely on the verbal information provided is an ongoing debate within cognitive psychology. There are cases where people only use base rates to reason or do something in between. If you're interested in learning more, please reach out to Joe. He likes this sort of thing and has thought and read about it a lot.

we frequently use an *indicator function*, $I(\cdot)$, which is 1 when the argument to the function I is true and zero when the argument is false. Thus, the probability density function for a continuous random variable uniformly distributed between 0 and 0.5 is defined as $p(x) = 2I(0 \leq x \leq 0.5)$. Note that the cumulative probability function for a continuous random variable X that is uniformly distributed between 0 and 0.5 is

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 2x & \text{for } 0 \leq x < 0.5 \\ 1 & \text{for } x \geq 1 \end{cases}$$

16. *Gaussians.* Although we will work with many distributions throughout the semester, one of the most common ones is the *Gaussian distribution* (also called the Normal distribution). The *Gaussian distribution* has two parameters: its mean μ, which is the center of the distribution (its expected value), and its variance σ^2, which is its spread. The probability of x under a *Gaussian distribution* is

$$p(x|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{\pi}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}.$$

In generative process notation, this is written as $x|\mu, \sigma^2 \sim N(\mu, \sigma^2)$, which is read as "the probability of x conditioned on μ and σ^2 is a Normal distribution with parameters μ and σ^2. Figure 5 illustrates the Gaussian distribution for $\sigma^2 = 0.5$ and $\sigma^2 = 1$. One quantity of interest is how to update our

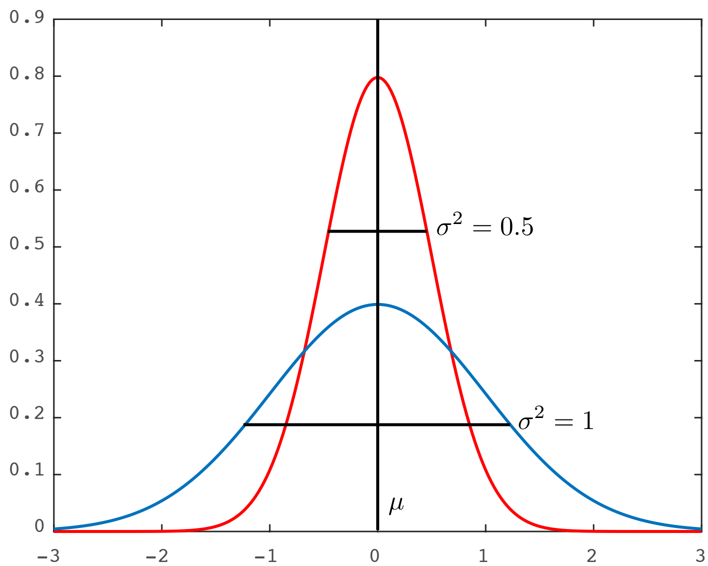

Figure 5: Gaussian probability density function for $\mu = 0$ with $\sigma^2 = 0.5$ (red) and $\sigma^2 = 1$ (blue).

beliefs in the mean of a Gaussian after observing one sample from it (and with some prior beliefs as to what the mean is). One generative process capturing this model is

$$\mu \sim N(\mu_0, \sigma_0^2) \qquad\qquad x|\mu, \sigma_x^2 \sim N(\mu, \sigma_x^2) \tag{3}$$

Thus, we are looking for the posterior probability $p(\mu|x)$, which is given by Bayes' rule $p(\mu|x) = \frac{p(x|\mu)p(\mu)}{p(x)}$. One way to solve this is to calculate each term and solve for the posterior distribution explicitly. However, in many cases there is an easier way (and remember the even easier way is to look up the result on Wikipedia `https://en.wikipedia.org/wiki/Conjugate_prior`).

Note that $p(x)$ does not change with changing μ. It only affects $p(\mu|x)$ as multiplying the probability of μ by the same constant value. $p(x)$ is the value that normalizes $p(x|\mu)p(\mu)$ such that integrating over all possible values for μ is 1 (one of the things necessary for it to be a probability distribution). So, what we need to do is figure out what $\int p(x|\mu)p(\mu)d\mu$. Integrating is hard. So, we hope that $p(x|\mu)p(\mu)$ corresponds to the functional form of a familiar probability distribution. If that is the case, we can see what parameters that distribution must be so that the whole quantity results in a valid probability distribution.

Lets follow through with this for the example defined by Equation 3. We drop the integral because we can examine the product in proportional form.

$$\begin{aligned} p(\mu|x) \quad \underbrace{\propto}_{\text{means proportional to}} \quad p(x|\mu)p(\mu) =& \frac{1}{\sigma_x\sqrt{\pi}} \exp\left\{-\frac{1}{2\sigma_x^2}(x-\mu)^2\right\} \frac{1}{\sigma_0\sqrt{\pi}} \exp\left\{-\frac{1}{2\sigma_0^2}(\mu-\mu_0)^2\right\} \\ =& \frac{1}{\sigma_x\sigma_0\pi} \exp\left\{-\frac{1}{2\sigma_x^2}(x-\mu)^2 - \frac{1}{2\sigma_0^2}(\mu-\mu_0)^2\right\} \end{aligned}$$

Now there are two ways to proceed: (a) Isolate μ to the point where we can identify the parameters of the updated distribution or (b) solve for it explicitly. If (a) does not feel ok to you yet, stick with (b) for now.

(a) μ only occurs in the exponential portion, so we can drop everything outside of it as proportional to it. Thus,

$$\begin{aligned} p(\mu \mid x) \propto& \exp\left\{-\frac{1}{2\sigma_x^2}(x-\mu)^2 - \frac{1}{2\sigma_0^2}(\mu-\mu_0)^2\right\} \\ =& \exp\left\{-\frac{1}{2\sigma_x^2}\left(x^2 - 2x\mu + \mu^2\right) - \frac{1}{2\sigma_0^2}\left(\mu^2 - 2\mu\mu_0 + \mu_0^2\right)\right\} \end{aligned}$$

Now any term that does not include μ can be dropped because it ultimately is part of the normalization constant (remember that $e^{a+b} = e^a e^b$). After dropping, we combine and write in terms of μ Thus,

$$\begin{aligned} p(\mu \mid x) \propto& \exp\left\{-\frac{1}{2\sigma_x^2}\left(2x\mu + \mu^2\right) - \frac{1}{2\sigma_0^2}\left(\mu^2 - 2\mu\mu_0\right)\right\} \\ =\propto& \exp\left\{-\mu^2\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_0^2}\right) + 2\mu\left(\frac{x}{2\sigma_x^2} + \frac{\mu_0}{2\sigma_0^2}\right)\right\} \end{aligned}$$

This looks like part of an expanded quadratic of a Gaussian (the missing parts are those we dropped during normalization). Whatever the coefficient is outside the quadratic will end up determining the updated variance. We first divide out the coefficient on μ^2 so that we have it outside the quadratic and can derive the updated variance from it. The updated mean will be the subtracted value in the quadratic. The value we have subtracting μ^2 right now is 2 times μ times the constant term of the quadratic. That constant term is the updated mean and so we solve for it second. So (first rewriting the last line),

$$\begin{aligned} p(\mu \mid x) \propto& \exp\left\{-\mu^2\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_0^2}\right) + 2\mu\left(\frac{x}{2\sigma_x^2} + \frac{\mu_0}{2\sigma_0^2}\right)\right\} \\ =& \exp\left\{-\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_0^2}\right)\left(\mu^2 - 2\mu\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_0^2}\right)^{-1}\left(\frac{x}{2\sigma_x^2} + \frac{\mu_0}{2\sigma_0^2}\right)\right)\right\} \end{aligned}$$

Denoting the updated mean and variance parameters as μ_1 and σ_1^2 (the subscript refers to having updated by 1 data point). We'll do the variance first.

$$\begin{aligned} \frac{1}{2\sigma_1^2} =& \left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_0^2}\right) \\ \Rightarrow \sigma_1^2 =& \left(\frac{1}{\sigma_x^2} + \frac{1}{\sigma_0^2}\right)^{-1} \end{aligned}$$

Remember that inside of the exponential, the updated Gaussian should look something like this: $-\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2 = -\frac{1}{2\sigma_1^2}(\mu - 2\mu\mu_1 + \mu_1^2)$. So, μ_1 is

$$\Rightarrow \mu_1 = \left(\frac{1}{2\sigma_x^2}+\frac{1}{2\sigma_0^2}\right)^{-1}\left(\frac{x}{2\sigma_x^2}+\frac{\mu_0}{2\sigma_0^2}\right) = \left(\frac{1}{\sigma_x^2}+\frac{1}{\sigma_0^2}\right)^{-1}\left(\frac{x}{\sigma_x^2}+\frac{\mu_0}{\sigma_0^2}\right)$$

(b) We will focus on the exponentiated terms because these are the only ones that depend on μ (and thus the rest are part of the normalizing constant).

$$\begin{aligned}&-\frac{1}{2\sigma_x^2}(x-\mu)^2 - \frac{1}{2\sigma_0^2}(\mu-\mu_0)^2 = -\frac{\sigma_0^2}{2\sigma_x^2\sigma_0^2}(x-\mu)^2 - \frac{\sigma_x^2}{2\sigma_0^2\sigma_x^2}(\mu-\mu_0)^2\\ &-\frac{1}{2\sigma_0^2\sigma_x^2}\left[\sigma_0^2\left(x^2-2\mu x+\mu^2\right)+\sigma_x^2\left(\mu^2-2\mu\mu_0+\mu_0^2\right)\right]\end{aligned}$$

Next we combine terms and look for some way of writing this equation as a function of μ and hope that there is a corresponding distribution that is that function exponentiated.

$$\begin{aligned}&-\frac{1}{2\sigma_0^2\sigma_x^2}\left[\sigma_0^2\left(x^2-2\mu x+\mu^2\right)+\sigma_x^2\left(\mu^2-2\mu\mu_0+\mu_0^2\right)\right]\\ =&-\frac{1}{2\sigma_0^2\sigma_x^2}\left[\mu^2(\sigma_0^2+\sigma_x^2)-2\mu(x\sigma_0^2+\mu_0\sigma_x^2)+\sigma_0^2x^2+\sigma_x^2\mu_0^2\right]\end{aligned}$$

The latter two sums can be brought out of the exponential (remember $e^{a+b}=e^ae^b$) and they can be dropped because the do not depend on μ (and so they are part of the constant). Dropping them and continuing on

$$\begin{aligned}&-\frac{1}{2\sigma_0^2\sigma_x^2}\left[\mu^2(\sigma_0^2+\sigma_x^2)-2\mu(x\sigma_0^2+\mu_0\sigma_x^2)+\sigma_0^2x^2+\sigma_x^2\mu_0^2\right] \propto -\frac{1}{2\sigma_0^2\sigma_x^2}\left[\mu^2(\sigma_0^2+\sigma_x^2)-2\mu(x\sigma_0^2+\mu_0\sigma_x^2)\right]\\ =&-\frac{\sigma_0^2+\sigma_x^2}{2\sigma_0^2\sigma_x^2}\left[\mu^2-2\mu\frac{x\sigma_0^2+\mu_0\sigma_x^2}{\sigma_0^2+\sigma_x^2}\right]\end{aligned}$$

Note that although this looks complicated, it is of the form $(\mu-c)^2$, which expands to $\mu^2 - 2\mu c + c^2$. Thus, we can reduce the above equation to

$$p(\mu|x) \propto \exp\left\{-\frac{\sigma_0^2+\sigma_x^2}{2\sigma_0^2\sigma_x^2}\left(\mu-\frac{x\sigma_0^2+\mu_0\sigma_x^2}{\sigma_0^2+\sigma_x^2}\right)^2\right\} \tag{4}$$

This looks like a Normal distribution with mean $\frac{\sigma_0^2+\mu_0\sigma_x^2}{\sigma_0^2+\sigma_x^2}x$ and variance $\left(\frac{\sigma_0^2+\sigma_x^2}{\sigma_0^2\sigma_x^2}\right)^{-1}$. This is the same result as Wikipedia gives, though they write it differently. Their update is that the mean is $\left(\mu_0/\sigma_0^2 + x/\sigma_x^2\right)/(1/\sigma_0^2+1/\sigma_x^2)$ ($n=1$ because we have only one observation). To see that our result is equivalent, first we will derive our variance to look like theirs:

$$\left(\frac{\sigma_0^2+\sigma_x^2}{\sigma_0^2\sigma_x^2}\right)^{-1} = \left(\frac{\sigma_0^2}{\sigma_0^2\sigma_x^2}+\frac{\sigma_x^2}{\sigma_0^2\sigma_x^2}\right)^{-1} = \left(\frac{1}{\sigma_x^2}+\frac{1}{\sigma_0^2}\right)^{-1} \tag{5}$$

The inverses of the variances add (this is why sometimes people parameterize using a "precision" parameter, which is the inverse variance.). We can do the same for the mean (first we pull out the inverse of our variance and convert that into their form):

$$\frac{x\sigma_0^2+\mu_0\sigma_x^2}{\sigma_0^2+\sigma_x^2} = \frac{\frac{\sigma_0^2\sigma_x^2}{\sigma_0^2\sigma_x^2}x\sigma_0^2+\mu_0\sigma_x^2}{\sigma_0^2+\sigma_x^2} = \sigma_0^2\sigma_x^2\frac{x/\sigma_x^2+\mu_0/\sigma_0^2}{\sigma_0^2+\sigma_x^2} = \frac{\sigma_0^2\sigma_x^2}{\sigma_0^2+\sigma_x^2}\left(x/\sigma_x^2+\mu_0/\sigma_0^2\right) \tag{6}$$

17. *Gaussians: Predictive update.* In some cases, we get a set of observations $X_1, \dots, X_N$ that are all generated from the same, unknown mean μ, and want to use the observed set to predict the next observation X_{N+1} from the unknown mean. We're going to be in the same generative process as before: $X_1, \dots, X_{N+1}|\mu, \sigma_x^2 \overset{iid}{\sim} N(\mu, \sigma_x^2)$, and $\mu|\mu_0, \sigma_0^2 \sim N(\mu_0, \sigma_0^2)$. *iid* means that the random variables are independent and identically distributed. We're going to assume that σ_x^2, μ_0, and σ_0^2 are known.[9] We are also going to use the posterior update for μ given N observations from Wikipedia $\mu|X_1, \dots, X_N$:

$$\mu|x_1, \dots, x_N \sim \mathrm{N}\left(\frac{\mu_0\sigma_0^{-2} + \sigma_x^{-2}\sum_{n=1}^N x_n}{\sigma_0^{-2} + N\sigma_x^{-2}}, \left[\sigma_0^{-2} + N\sigma_x^{-2}\right]^{-1}\right)$$

Lets derive the predictive update using this information and the sum rule.

$$\begin{aligned}
p(X_{N+1} \mid X_1, \dots, X_N) &= \int p(X_{N+1}, \mu \mid X_1, \dots, X_N, \sigma_x^2) d\mu \\
&= \int p(X_{N+1} \mid \mu, X_1, \dots, X_N, \sigma_x^2) p(\mu \mid X_1, \dots, X_N) d\mu \\
&= \int p(X_{N+1} \mid \mu, \sigma_x^2) p(\mu \mid X_1, \dots, X_N) d\mu
\end{aligned}$$

Because this is getting a bit unwieldy, lets define variables for the posterior updates: σ_N^2 and μ_N.

$$\mu_N = \frac{\mu_0\sigma_0^{-2} + \sigma_x^{-2}\sum_{n=1}^N x_n}{\sigma_0^{-2} + N\sigma_x^{-2}} \qquad\qquad \sigma_N = \left[\sigma_0^{-2} + N\sigma_x^{-2}\right]^{-1}$$

So,

$$\begin{aligned}
&p(X_{N+1} \mid X_1, \dots, X_N) = \frac{1}{\sigma_N\sigma_x\pi}\int \exp\left\{-\frac{1}{2\sigma_x^2}\left(X_{N+1} - \mu\right)^2\right\}\exp\left\{-\frac{1}{2\sigma_N^2}\left(\mu - \mu_N\right)^2\right\} d\mu \\
&= \frac{1}{\sigma_N\sigma_x\pi}\int \exp\left\{-\frac{1}{2\sigma_x^2}\left(X_{N+1}^2 - 2X_{N+1}\mu + \mu^2\right) - \frac{1}{2\sigma_N^2}\left(\mu^2 - 2\mu_N\mu + \mu_N^2\right)\right\} d\mu \\
&= \frac{1}{\sigma_N\sigma_x\pi}\exp\left\{-\frac{1}{2\sigma_x^2}X_{N+1}^2 - \frac{1}{2\sigma_N^2}\mu_N^2\right\}\int\left(\exp\left\{-\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_N^2}\right)\mu^2 + 2\left(\frac{1}{2\sigma_x^2}X_{N+1} + \frac{1}{2\sigma_N^2}\mu_N\right)\mu\right\}\right) d\mu
\end{aligned}$$

The integral has the (incomplete) form of an unnormalized normal distribution. To solve it, we need to complete it (called "completing the square"). Once we have the completed form, we know it must integrate to whatever the standard normalizing constant is for that unnormalized Normal distribution (because that's the constant that makes sure it sums to 1!). Focusing on the exponential term in the integral:

$$\begin{aligned}
&-\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_N^2}\right)\mu^2 + 2\left(\frac{1}{2\sigma_x^2}X_{N+1} + \frac{1}{2\sigma_N^2}\mu_N\right)\mu = \\
&-\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_N^2}\right)\left(\mu^2 - 2\left(\frac{1}{2\sigma_x^2}X_{N+1} + \frac{1}{2\sigma_N^2}\mu_N\right)\left(\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_N^2}\right)^{-1}\mu\right)
\end{aligned}$$

Note that

$$\frac{1}{2\sigma_x^2} + \frac{1}{2\sigma_N^2} = \frac{\sigma_N^2}{2\sigma_x^2\sigma_N^2} + \frac{\sigma_x^2}{2\sigma_N^2\sigma_x^2} = \frac{\sigma_x^2 + \sigma_N^2}{2\sigma_x^2\sigma_N^2}$$

and

$$\frac{1}{2\sigma_x^2}X_{N+1} + \frac{1}{2\sigma_N^2}\mu_N = \frac{\sigma_N^2 X_{N+1}}{2\sigma_x^2\sigma_N^2} + \frac{\sigma_x^2\mu_N}{2\sigma_N^2\sigma_x^2} = \frac{\sigma_x^2\mu_N + \sigma_N^2 X_{N+1}}{2\sigma_x^2\sigma_N^2}$$

[9] If you are concerned that we are always assuming known parameter values, deriving the full updates when all parameters of the likelihood are unknown is doing the same process, but for multiple steps.

Using that to simplify and continue:

$$
\begin{aligned}
&-\left(\frac{\sigma_x^2+\sigma_N^2}{2\sigma_x^2\sigma_N^2}\right)\left(\mu^2-2\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{2\sigma_x^2\sigma_N^2}\frac{2\sigma_x^2\sigma_N^2}{\sigma_x^2+\sigma_N^2}\mu\right)\\
&-\left(\frac{\sigma_x^2+\sigma_N^2}{2\sigma_x^2\sigma_N^2}\right)\left(\mu^2-2\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{\sigma_x^2+\sigma_N^2}\mu\right)
\end{aligned}
$$

This means that the coefficient of μ (not including 2) is the "mean" of the Normal distribution of the integral. But, we can't just complete it, and make it proportional without keeping track of the term because the needed term includes X_{N+1}. We know it is missing the $+c$ term of the quadratic, which is the coefficient of μ^2 (multiplied by the outer variance term). So, to complete the square, we will be (implicitly) including

$$
-\left(\frac{\sigma_x^2+\sigma_N^2}{2\sigma_x^2\sigma_N^2}\right)\left(\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{\sigma_x^2+\sigma_N^2}\right)^2
$$

Thus, to make sure our net effect on the equation is still 1, we need to include its inverse (don't forget we're inside of an exponential too!). So we will include this term outside of the integral:

$$
\exp\left\{\left(\frac{\sigma_x^2+\sigma_N^2}{2\sigma_x^2\sigma_N^2}\right)\left(\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{\sigma_x^2+\sigma_N^2}\right)^2\right\}
$$

Completing the square, we get the following inside the exponential inside of the integral.

$$
-\left(\frac{\sigma_x^2+\sigma_N^2}{2\sigma_x^2\sigma_N^2}\right)\left(\mu-\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{\sigma_x^2+\sigma_N^2}\right)^2
$$

This is a Normal distribution with mean $\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{\sigma_x^2+\sigma_N^2}$ and variance $\frac{\sigma_x^2\sigma_N^2}{\sigma_x^2+\sigma_N^2}$. We know that the Normal distribution integrates to 1 (this has to be true for it to be a probability density). Thus,

$$
\begin{aligned}
&\frac{1}{\sigma\sqrt{2\pi}}\int\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}dx=1\\
\Rightarrow&\int\exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}dx=\sigma\sqrt{2\pi}
\end{aligned}
$$

So our integral must equal $\left(\frac{\sigma_x^2\sigma_N^2}{\sigma_x^2+\sigma_N^2}\right)^{1/2}\sqrt{2\pi}$.

Putting this all together:

$$
\begin{aligned}
&=\frac{1}{\sigma_N\sigma_x\pi}\exp\left\{-\frac{1}{2\sigma_x^2}X_{N+1}^2-\frac{1}{2\sigma_N^2}\mu_N^2\right\}\exp\left\{\left(\frac{\sigma_x^2+\sigma_N^2}{2\sigma_x^2\sigma_N^2}\right)\left(\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{\sigma_x^2+\sigma_N^2}\right)^2\right\}\left(\frac{\sigma_x^2\sigma_N^2}{\sigma_x^2+\sigma_N^2}\right)^{1/2}\sqrt{2\pi}\\
&=\frac{1}{\sqrt{2\pi(\sigma_x^2+\sigma_N^2)}}\exp\left\{-\frac{1}{2\sigma_x^2}X_{N+1}^2-\frac{1}{2\sigma_N^2}\mu_N^2+\left(\frac{\sigma_x^2+\sigma_N^2}{2\sigma_x^2\sigma_N^2}\right)\left(\frac{\sigma_x^2\mu_N+\sigma_N^2X_{N+1}}{\sigma_x^2+\sigma_N^2}\right)^2\right\}
\end{aligned}
$$

Note that from outside of the exponential, we already have the variance! (It's $\sigma_x^2+\sigma_N^2$). If we continue the manipulation, we would get the following solution:

$$
x_{N+1}|x_1,\ldots,x_N\sim \mathrm{N}\left(\frac{\mu_0\sigma_0^{-2}+\sigma_x^{-2}\sum_{n=1}^N x_n}{\sigma_0^{-2}+N\sigma_x^{-2}},\left[\sigma_0^{-2}+N\sigma_x^{-2}\right]^{-1}+\sigma_x^2\right)
$$

18. *Expectations.* Sometimes we are interested in what values we should expect from a random variable (or a function of a random variable). For example, we are often interested in the average. The *expected value* of a random variable X is $E[X] = \sum_x xP(X = x)$ for discrete random variables and $E[X] = \int xp(x)dx$ for continuous random variables. Note that sometimes we're interested in expected value of a function $f(x)$ a random variable (for example, the variance), which is $E[f(X)] = \sum_x f(x)P(X = x)$ for discrete random variables and $E[f(X)] = \int f(x)p(x)dx$. One neat trick is to represent probabilities in terms of the expectation of indicator functions. For example, the probability that a continuous random variable is between 1 and 3 can be represented as $E[I(1 \leq X \leq 3)]$. This will be used often when we discuss approximations. Here are some common expectations and properties:

    (a) $E[X]$ is the mean (or average) of a random variable.

    (b) $E[X^k]$ is sometimes called the k-th moment of random variable X.

    (c) $E[X^2] - (E[X])^2$ is the variance of a random variable.

    (d) *Linearity.* $E[\alpha X + \beta Y] = \alpha E[X] + \beta E[Y]$ for random variables X and Y, and constants α and β.

    (e) You can also have conditional expectations (i.e., the expected value of X given Y), which is defined in the same way and has the same properties as standard expectations except using $P(X|Y)$.

    (f) Note that if $X \sim N(\mu, \sigma^2)$, then $E[X] = \mu$.

19. *The Dirichlet-Multinomial Model* (the generalization of the Beta-Binomial Model). A common model for discrete (nominal) data is the Dirichlet-Multinomial model. According to this model, each data point is discrete-valued and takes each value with probability given by its corresponding element in a vector of probabilities $\vec{\theta}$. If there are K possible discrete values, $\vec{\theta}$ has length K and forms a probability distribution. This means that for each k, $\theta_k \geq 0$, and the sum of the elements of $\vec{\theta}$ is one – $\sum_{k=1}^{K} \theta_k = 1$. So, given a probability distribution over discrete-values, the data are multinomial distributed: $\vec{x}|\vec{\theta} \sim \text{Multinomial}(\vec{\theta})$. Lets say there were N total outcomes and x_k is the number of outcomes with value k. The form of the distribution is

$$P(\vec{x}|\vec{\theta}) = \frac{N!}{x_1! \dots x_K!} \theta_1^{x_1} \dots \theta_K^{x_K} = \frac{N!}{\prod_k x_k!} \prod_{k=1}^{K} \theta_k^{x_k}$$

    The conjugate prior for the Multinomial distribution is the Dirichlet distribution (the generalization of the Beta distribution). It is a probability distribution over discrete-valued probability distributions (aka $\vec{\theta}$). It has a parameter α_k, corresponding to each element of $\vec{\theta}$. The constraint on the parameters is $\alpha_k > 0$. Sometimes α is a scalar, which is equivalent to setting each element of the vector to α. The Dirichlet probability distribution is

$$p(\vec{\theta}|\vec{\alpha}) = \frac{\Gamma\left(\sum_k \alpha_k\right)}{\prod_k \Gamma(\alpha_k)} \prod_k \theta_k^{\alpha_k - 1}.$$

    So, the Dirichlet-multinomial model written as a generative process is $\vec{\theta}|\vec{\alpha} \sim \text{Dirichlet}(\alpha)$ and $\vec{x}|\vec{\theta} \sim \text{Multinomial}(\vec{\theta})$.

20. Posterior update for *the Dirichlet-Multinomial Model* and expected value.

$$\begin{aligned} p(\vec{\theta}|\vec{x}, \vec{\alpha}) &\propto P(\vec{x}|\vec{\theta})p(\vec{\theta}|\vec{\alpha}) \\ &\propto \prod_k \theta_k^{x_k} \prod_k \theta_k^{\alpha_k - 1} = \prod_k \theta_k^{x_k + \alpha_k - 1} \end{aligned}$$

This is the functional form for a Dirichlet distribution with parameters $\vec{x} + \vec{\alpha}$.

What is the expected proportion of k-valued outcomes? This is $E[\theta_k]$.

$$\begin{aligned} E[\theta_k] &= \int d\vec{\theta}\theta_k p(\vec{\theta}) = \int d\vec{\theta}\theta_k \frac{\Gamma\left(\sum_i \alpha_i\right)}{\prod_i \Gamma(\alpha_i)} \prod_i \theta_i^{\alpha_i - 1} \\ &= \frac{\Gamma\left(\sum_i \alpha_i\right)}{\prod_i \Gamma(\alpha_i)} \int d\vec{\theta}\theta_k^{\alpha_k} \prod_{i \neq k} \theta_i^{\alpha - 1} \end{aligned}$$

The integral is of an unnormalized Dirichlet distribution with the same parameter $\vec{\alpha}$ but with one added to the k-th element. Thus, we continue with plugging in the inverted normalization constant for this Dirichlet distribution for the integral.

$$\begin{aligned} E[\theta_k] &= \frac{\Gamma\left(\sum_i \alpha_i\right)}{\prod_i \Gamma(\alpha_i)} \frac{\Gamma(\alpha_k + 1) \prod_{i \neq k} \Gamma(\alpha_i)}{\Gamma\left(1 + \sum_i \alpha_i\right)} = \frac{\Gamma\left(\sum_i \alpha_i\right) \Gamma(\alpha_k + 1) \prod_{i \neq k} \Gamma(\alpha_i)}{\Gamma\left(1 + \sum_i \alpha_i\right) \prod_i \Gamma(\alpha_i)} \\ &= \frac{\Gamma\left(\sum_i \alpha_i\right) \alpha_k \Gamma(\alpha_k) \prod_{i \neq k} \Gamma(\alpha_i)}{\left(\sum_i \alpha_i\right) \Gamma(\sum_i \alpha_i) \prod_i \Gamma_i} = \frac{\Gamma\left(\sum_i \alpha_i\right) \alpha_k \prod_i \Gamma(\alpha_i)}{\left(\sum_i \alpha_i\right) \Gamma(\sum_i \alpha_i) \prod_i \Gamma_i} = \frac{\alpha_k}{\Gamma(\sum_i \alpha_i)} \end{aligned}$$

Glossary

conditional probability The probability that event X occurs given event Y occurred is equivalent to given that we are in the worlds where Y is true, how many of those worlds is X true.

Consistent Belief updating systems have a set of rules that tell you how to go from one set of beliefs to a new set of beliefs. In a consistent system, the new set of beliefs is always the same no matter what order or way you apply the set of rules in the system.

countable Any finite set is countable. Infinite sets are countable if it has one-to-one correspondence with the natural numbers.

cumulative distribution function The cumulative distribution function $F(x)$ provides the probability of observing a value less than or equal to x, which is defined to be $F(x) = \int_{-\infty}^{x} p(t)dt$.

expected value The expected value of a random variable X is $E[X] = \sum_x xP(X = x)$ for discrete random variables and $E[X] = \int xp(x)dx$ for continuous random variables. Note that sometimes we're interested in expected value of a function $f(x)$ a random variable (for example, the variance), which is $E[f(X)] = \sum_x f(x)P(X = x)$ for discrete random variables and $E[f(X)] = \int f(x)p(x)dx$. One neat trick is to represent probabilities in terms of the expectation of indicator functions. For example, the probability that a continuous random variable is between 1 and 3 can be represented as $E[I(1 \leq X \leq 3)]$. This will be used often when we discuss approximations.

Gaussian distribution The Gaussian distribution (also known as the Normal distribution) is a continuous distribution defined over the real numbers. The one-dimensional (univariate) case is defined via two parameters: a mean μ and variance σ^2. The probability of x under a Gaussian distribution with mean μ and variance σ^2 is $p(x|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-0.5\sigma^{-2}\left(x - \mu\right)^2\right\}$.

independent Two events are independent when their joint probability is equal to the product of their individual probabilities. So, if A and B are random variables, they are independent if and only if $P(A, B) = P(A)P(B)$. Equivalently, it is the case when $P(A|B) = P(A)$ or $P(B|A) = P(B)$.

indicator function An indicator function $I(\cdot)$ is 1 when the argument to I is true and 0 when the argument is false. For example, $I(0 < x < 1)$ is 1 when x is between 0 and 1, and zero for all other values of x.

joint probability The joint probability of two events is the number of outcomes where both events occur out of the number of possible outcomes.

mutually exclusive A set of hypotheses where only one hypothesis can be true at a time (in a possible world). In other words, if we treat hypotheses as events (a set of atomic probability units), mutually exclusive means that the intersection of any two hypotheses is null or empty.

probability density function A probability density function defines the relative likelihood at each x. Due to the continuous space, the "probability" at any particular x is 0 (as there are an infinite, continuous space of possibilities). Instead of examining probabilities of any particular x, we examine the probability of a range of values. The probability of a region of values according to a probability density function is its area, or integral, over the probability density function.

set A set is a collection of elements or members.

set size The size of a set (or its cardinality) is the number of elements in the set. The size of set A is often written as $|A|$.

sum rule The sum rule is that the probability that a random variable A takes some value a, $P(A = a)$, is the same as summing the joint probability that $A = a$ and $B = b$ over all the possible values b that the second variable B can take.

A Why use Bayesian inference as a normative theory of how to update one's belief in hypotheses after observing data?

Note: We will denote someone's belief using a function $P(\cdot)$, which reads as "the probability of " (and then whatever is in the parentheses). As we are discussing people's belief in hypotheses, we will write that as $P(h)$, which is the probability of h or the belief someone has in a particular hypothesis h.

1. Cox's theorem. If you accept the following axioms, then you should update your belief in each hypothesis h about the state of the world according to Bayes' rule (to stay *Consistent*):

   (a) Hypotheses cannot have negative probability: $P(h) \geq 0$ for every h.

   (b) There is some hypothesis that is true. $P(\text{some } h \text{ is true}) = 1$.

   (c) If we have a *countable* set of *mutually exclusive* hypotheses $H_n = \{h_1, \ldots, h_n\}$, $P(\text{some } h \in H_n \text{ is true}) = \sum_i^n P(h_i)$

   In sum, accepting these axioms is akin to accepting that beliefs about the state of the world can be encoded as probability distributions (Otherwise it will lead to inconsistent beliefs).[10]

[10] As you may suspect there is a literature debating that *these* are the correct assumptions to have in philosophy. However, this will be beyond the scope of the class. Most machine learning and many artificial intelligence theorists accept Bayesian theory as the normative theory for induction.

2. Dutch books. Assume that a person will *always* accept a "fair" bet or better — one whose expected financial is gain greater than or equal to zero. For example, Jane believes that there is a $2/3$ probability that Joe will show 30 or more than cat pictures by the end of the semester and a $1/3$ probability that he will show fewer than 30. She should be willing to take a bet where she loses \$1 if Joe shows 30 or more cat pictures and gets \$2 if he shows fewer. Assume there are different events in the world that could effect the odds (e.g., Joe gets sick and cannot come to class one day). If she does not assign probabilities and take bets according to Bayesian inference, it is possible to offer a combination of bets such that Jane will accept each individual bet, but no matter what (i.e., with probability 1), there is a sequence of bets where Jane is *guaranteed* to lose money! See the Wikipedia (`https://en.wikipedia.org/wiki/Dutch_book`) on Dutch books for a nice example of such a case.